TP 1: Returns and Random Walks

Statistics and Data Analysis for Financial Engineering, Chapters 1 and 2

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Introduction

The goal of this practical work is to study the basic properties of financial returns and to test the model most often used for prices, the geometric (lognormal) random walk. In this model the daily log returns are independent and normally distributed with a constant mean μ and a constant standard deviation σ. Eight exercises look at this model from different sides: looking at the data, checking normality, checking constant variance, converting between types of returns, dividends, simulation, a leverage problem solved by Monte Carlo, and finally formal statistical tests.

All the calculations were done with R. The data come from the R packages `datasets` (the European stock indices `EuStockMarkets`) and `Ecdat` (`SP500`, `CRSPday` and `Garch`). Exercise 5 uses prices downloaded from Yahoo Finance and was run with an internet connection. In this report the code is not shown. For each exercise I explain the objective, how the result is obtained, the numerical results, and what the graphs show.

One remark on dates. The data sets `EuStockMarkets` and `SP500` do not contain calendar dates, only the order of the observations (and, for the European indices, a time stamp in fractional years with 260 trading days per year). The dates given below are therefore found from this time stamp and from the known history of the markets, and they should be read as approximate.

1 Exercise 1. Looking at the data (EuStockMarkets)

(a) The four indices

Objective. The first principle of Chapter 1 is to always look at the data before modelling. The aim is to see the general trend of each index and the periods where it moves the most.

Data and method. The data set contains 1860 daily closing values of the DAX (Germany), SMI (Switzerland), CAC (France) and FTSE (United Kingdom), from mid-1991 to August 1998. The four series are plotted against time. To make "moves the most" precise, I also computed the annualised volatility of the daily log returns year by year, and the standard deviation over a rolling window of 60 days.

Results. Figure 1 shows that all four indices rise over the period, but not at the same pace. Between the first and the last observation the DAX goes from 1629 to 5474 (+236%), the SMI from 1678 to 7676 (+357%), the CAC from 1773 to 3995 (+125%) and the FTSE from 2444

to 5455 (+123%). The SMI is therefore the strongest index and the CAC and the FTSE the weakest.

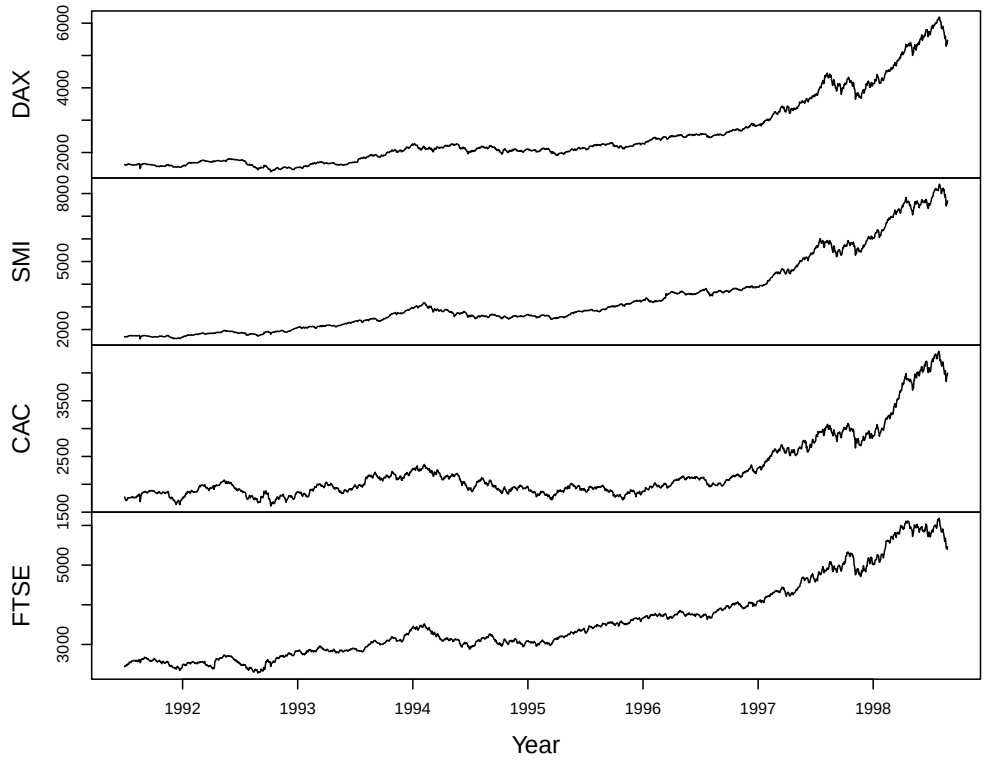


Figure 1: The four European indices, 1991–1998.

Explanation of the graph. The four panels have a similar shape. From 1991 to 1996 the indices move sideways or rise slowly: the DAX, for example, stays between about 1400 and 2500. All four reach a local peak at the beginning of 1994 (the SMI and the FTSE most clearly) and then decline during 1994, before recovering. From 1996 the rise becomes strong and steeper in 1997, with a visible dip in the autumn of 1997, followed by a new rise to the maximum of the sample in mid-1998 (DAX 6186, SMI 8412, CAC 4388, FTSE 6179) and a sharp fall at the very end of the series. The rise of 1996–1998 is the most visible feature, and the largest swings occur in 1997 and 1998. This is confirmed by the annualised volatility of the DAX, which is about 11% in 1996 and 23% in 1997. The rolling volatility is highest at the end of 1997 for the DAX and the SMI, at the beginning of 1998 for the CAC, and in autumn 1992 for the FTSE. So the period where the markets move the most is 1997–1998 for the three continental indices, while the FTSE has its most turbulent moment earlier, in 1992.

(b) Log returns, scatter-plot matrix and correlations

Objective. To measure how closely the four markets move together from day to day.

Method. The daily log returns are computed as the difference of the logarithms of two consecutive closing values. The scatter-plot matrix shows every pair of returns, and the correlation matrix summarises it with one number per pair.

Results. The correlations of the daily log returns are:

	DAX	SMI	CAC	FTSE
DAX	1.000	0.703	0.734	0.639
SMI	0.703	1.000	0.616	0.585
CAC	0.734	0.616	1.000	0.649
FTSE	0.639	0.585	0.649	1.000

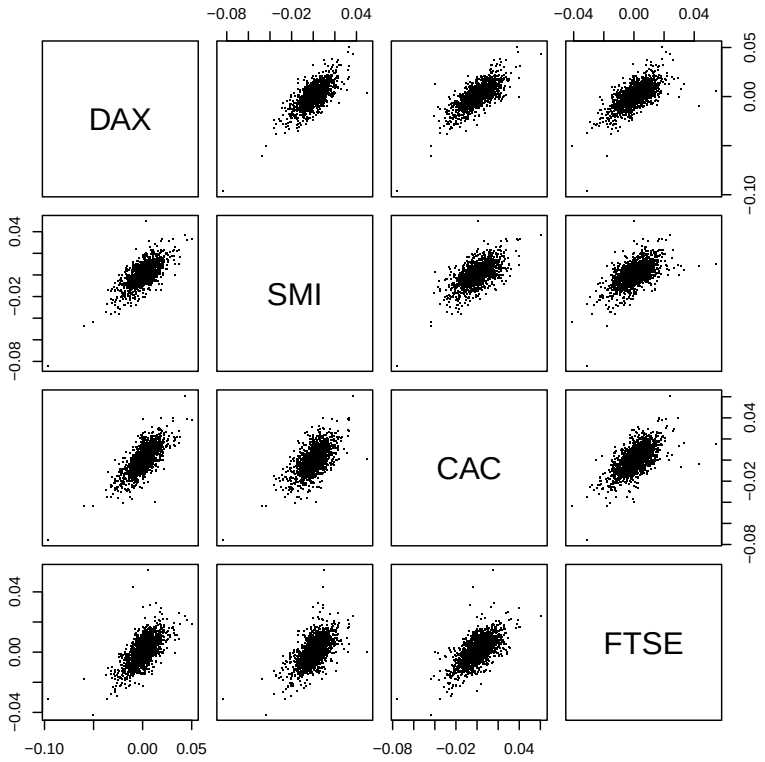


Figure 2: Scatter-plot matrix of the daily log returns.

Explanation of the graph. In Figure 2 each small panel plots the return of one index against the return of another. Every cloud is elongated along the diagonal going up, which is the sign of a positive correlation: when one market rises, the others tend to rise too. The cloud is thinnest (closest to a line) for DAX–CAC, which is the pair with the highest correlation, and it is the roundest for SMI–FTSE. A few points are far from the centre of each cloud; these are the days of large moves, discussed in part (c).

Conclusion. The two most correlated markets are the DAX and the CAC (0.734), the two largest continental economies. The least correlated market is the FTSE: its correlations with the other three are 0.639, 0.585 and 0.649 (average 0.624), slightly lower than the SMI (average 0.635), and the weakest pair is SMI–FTSE (0.585). It is natural that the British market, which is outside the continental group, moves a little more independently. It is also worth noting that the correlations of the index *levels* are much higher (above 0.91 for all pairs). This is misleading, because two series that both trend upward are strongly correlated even if their daily moves are unrelated. This is why returns, and not prices, must be used to measure dependence.

(c) Worst daily log returns

Objective. To find the most extreme day of each market and to see whether the markets suffer their worst day together.

Method. For each index the minimum of the daily log returns and its position in the sample are extracted, and the time stamp of that observation is read.

Results.

Index	Worst log return	Observation	Time stamp
DAX	-0.0963	35	1991.63
SMI	-0.0838	35	1991.63
CAC	-0.0758	35	1991.63
FTSE	-0.0414	330	1992.77

The DAX, the SMI and the CAC have the same worst day, observation 35. On the same day the FTSE also fell sharply (-0.0312), which is its second worst return in the whole sample. Only the FTSE has its worst day elsewhere, in autumn 1992.

Interpretation. A common worst day for three markets shows that a shock that hits Europe as a whole is transmitted immediately to all markets. Using the time stamp, observation 35 corresponds to about 19 August 1991, the day of the attempted coup against Mikhail Gorbachev in the Soviet Union, which created sudden uncertainty on the European markets, especially the German one. The FTSE minimum in autumn 1992 is close to the crisis of the European Monetary System and of sterling in September 1992. These links are plausible but are not proved by the data alone, since the series carry no dates.

(d) Long DAX, short FTSE

Objective. To compare the performance of two indices that have very different levels (the DAX around 1600–6000 and the FTSE around 2400–5500), using log returns only.

Method. An investor puts the same amount in the DAX (long) and sells the FTSE short. The log of the ratio of the value of the DAX investment to the value of the FTSE investment is the cumulative sum of the daily differences of the two log returns. It is plotted against time, with a dashed line at zero.

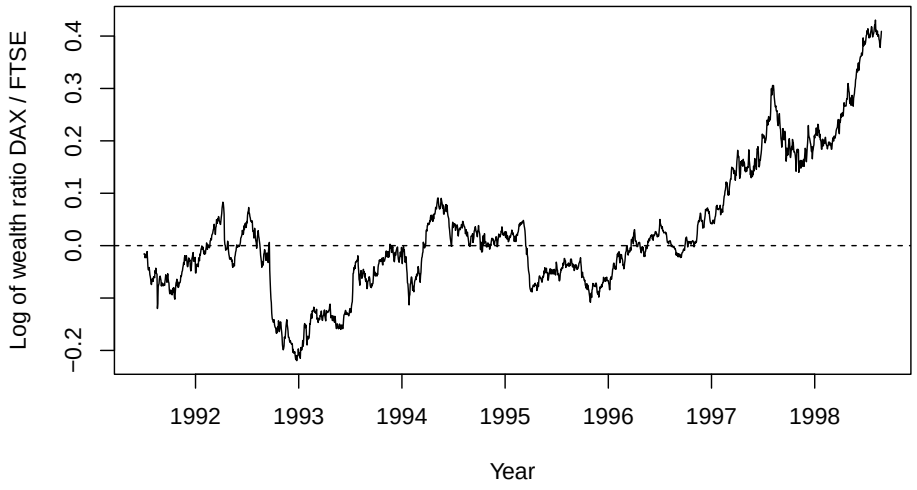


Figure 3: Log of the ratio of the DAX investment to the FTSE investment.

Explanation of the graph. When the curve in Figure 3 is above zero, the DAX investment is worth more than the FTSE one, and when it is below zero, the opposite. The curve starts at zero, goes negative in 1991–1992 and reaches its lowest value (-0.22) at the end of 1992, because the FTSE did better in that period. It then recovers, fluctuates, and rises strongly in 1996–1997 up to its maximum ($+0.43$) in mid-1998, finishing at $+0.41$.

Conclusion. The final value 0.41 means that the DAX investment is worth $e^{0.41} \approx 1.51$ times the FTSE investment at the end of the period. So yes, the DAX outperformed the FTSE over the whole period, even though it underperformed during the first years. This is only a comparison of two investments of equal size. It does not include transaction costs, the cost of borrowing the FTSE for the short sale, or the exchange rate between the mark and the pound.

2 Exercise 2. The normal distribution is not normal (SP500)

Objective. To show, with the S&P 500, that the normal model is not appropriate for extreme returns: very large losses happen far more often than the normal distribution allows.

Data. 2783 daily log returns of the S&P 500, from 1981 to 1991. The sample mean is 0.000418 and the sample standard deviation is 0.010863 (about 1.09% per day).

(a) The worst day

The minimum of the sample is at observation 1805. Its log return is -0.2280, which corresponds to a net return of $e^{-0.2280} - 1 = -20.39\%$. The sample has no dates, but a one-day fall of more than 20% is the Black Monday crash of 19 October 1987. The log return is more negative than the net return, which is a general property: a loss of 20.39% corresponds to a log return of -22.80%.

(b) Probability and waiting time under the normal model

Method. The worst return is standardised by subtracting the sample mean and dividing by the sample standard deviation. The normal distribution function gives the probability of a day at least this bad, and the inverse of this probability is the average waiting time in days, which is divided by 252 to give years.

Results. The standardised return is -21.03, which means that the crash is more than 21 standard deviations below the mean. Under the normal model the probability of such a day is 1.83×10^{-98}, and one would wait on average 5.5×10^{97} days, or 2.2×10^{95} years. For comparison, the age of the universe is about 1.4×10^{10} years. The normal model says that this day should never happen, yet it happened in a sample of eleven years.

(c) Number of extreme days

Method. The standardised returns are computed for the whole sample and the days with absolute standardised return above 3, 4 and 5 are counted. They are compared with the number expected under the normal model, which is the sample size multiplied by the normal probability of exceeding the threshold in absolute value.

Threshold	Observed	Expected (normal)	Expected (t_4)
3	20	7.51	36.8
4	11	0.176	13.4
5	6	0.0016	5.9

Interpretation. The observed counts are 2.7 times the normal prediction at 3 standard deviations, 62 times at 4 standard deviations, and thousands of times at 5 standard deviations. The farther in the tail, the worse the normal model is. The last column, added for comparison, uses the t-distribution with 4 degrees of freedom of part (d). It overestimates the number at the threshold 3 but is very close to the observed counts at 4 and 5, which shows that a heavy-tailed distribution describes the tails much better.

(d) The same probability with a t-distribution

Objective. To see how much the answer changes with a distribution that has heavier tails.

Method. A t-distribution with 4 degrees of freedom has variance $4/2 = 2$, so it is divided by $\sqrt{2}$ to have variance 1. The same standardised return (-21.03) is then evaluated with this distribution.

Results. The probability becomes 3.8×10^{-6}, and the average waiting time is about 262 600 days, or about 1040 years. This is still long, but it is a possible event, and it is larger than the normal result by a factor of about 10^{92}. The waiting time under the normal model is 10^{95} years and under the t_4 model about 10^3 years. The conclusion is that the exact probability of an extreme event is extremely sensitive to the tail of the distribution assumed, and that the normal model is dangerous for risk management.

(e) Normal QQ plot

Objective. To compare the whole distribution of the returns with the normal distribution graphically.

Method. The sorted returns (vertical axis) are plotted against the quantiles of a normal distribution (horizontal axis). If the data were normal, the points would lie on the straight line.

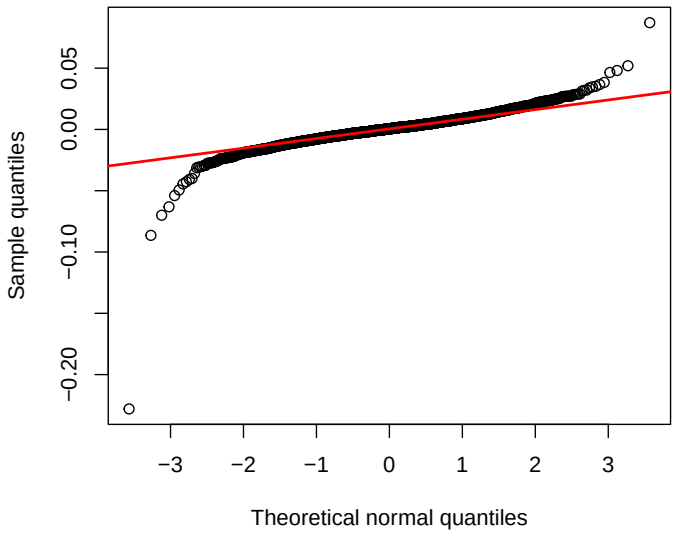


Figure 4: Normal QQ plot of the S&P 500 daily log returns.

Explanation of the graph. In Figure 4 the points in the centre, where most of the days are, follow the red line closely. In both tails the points leave the line: on the right (large gains) they are above it, and on the left (large losses) they are below it. The departure is much bigger on the left, and a single point (the crash of 1987) is far under the line at the bottom left. This S-shaped pattern is the signature of heavy tails (the sample kurtosis is about 77, against 3 for a normal distribution) and the left side is more extreme than the right (the skewness is -3.5), so crashes are bigger than rallies.

3 Exercise 3. Variances are not constant; measuring risk (CRSPday)

Data. 2528 daily net returns of GE, IBM, Mobil and the CRSP value-weighted index from 1989 to 1998, with about 252 observations per year.

(a) Annualised volatility by year

Objective. To check whether the standard deviation of returns is constant over time, as the random walk model assumes.

Method. For each asset and each calendar year, the standard deviation of the daily returns is

computed and multiplied by $\sqrt{252}$ to express it on a yearly scale.

Year	GE	IBM	Mobil	CRSP
1989	20.8%	17.0%	19.7%	10.9%
1990	25.0%	20.1%	20.3%	14.3%
1991	22.9%	24.0%	18.4%	12.9%
1992	16.5%	27.3%	19.7%	8.9%
1993	14.0%	30.0%	18.0%	8.0%
1994	19.2%	30.1%	15.4%	9.2%
1995	17.7%	23.3%	16.5%	7.4%
1996	20.1%	34.0%	19.3%	10.8%
1997	27.7%	35.4%	22.8%	15.8%
1998	29.0%	30.3%	31.1%	19.4%
All years	21.8%	27.8%	20.5%	12.3%

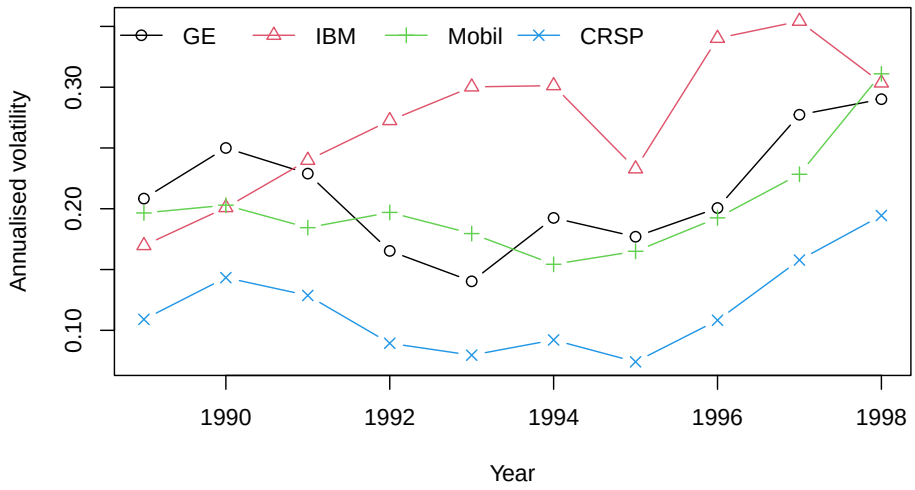


Figure 5: Annualised volatility of GE, IBM, Mobil and CRSP by year.

Explanation of the graph. Each line in Figure 5 would be flat if the volatility were constant. This is clearly not the case. The four curves move up and down, and they often move together: they are high in 1990, fall in 1992–1995, and rise sharply in 1997–1998. The market index CRSP varies from 7.4% in 1995 to 19.4% in 1998, so its volatility is multiplied by 2.6 in three years. The IBM curve has its own pattern, with a strong rise from 17% in 1989 to over 30% between 1993 and 1994 and a peak of 35.4% in 1997.

Conclusion. The volatility is not constant over time. IBM is the most volatile asset overall (27.8% a year) and in most years, with a maximum of 35.4% in 1997; the exceptions are 1989–1990 (GE) and 1998 (Mobil, at 31.1%). The index CRSP is always the least volatile, because it is a diversified portfolio in which the individual risks partly cancel out.

(b) Autocorrelation of IBM returns and squared returns

Objective. To test two things: whether today's return helps to predict tomorrow's return (independence), and whether large movements are followed by large movements (volatility clustering).

Method. The sample autocorrelation function (ACF) is computed for the IBM returns and for their squares. The dashed horizontal lines are the 95% bounds, here about ± 0.039, within which a correlation is not significantly different from zero.

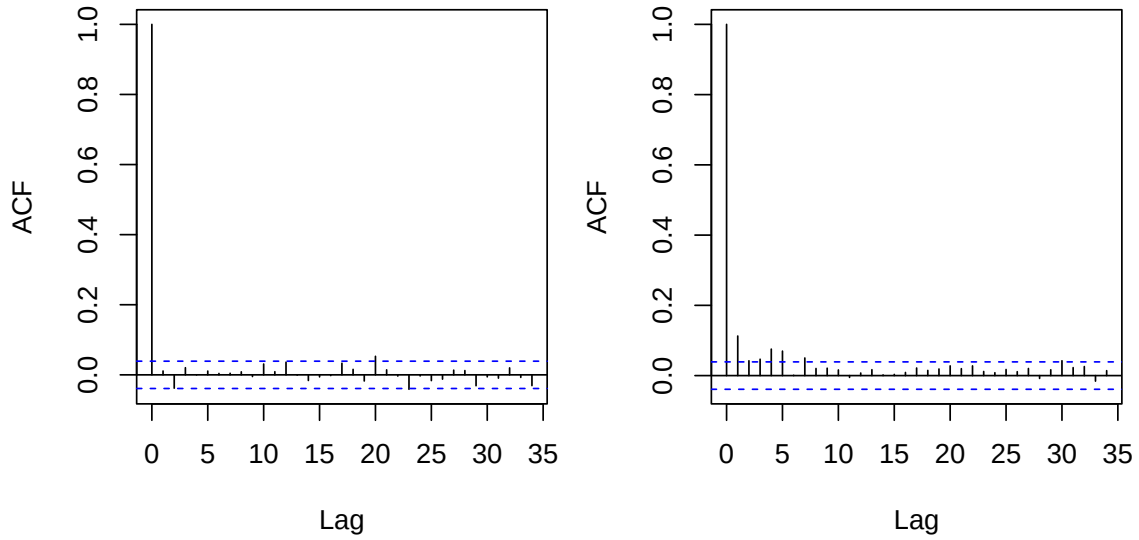


Figure 6: ACF of the IBM daily returns (left) and of their squares (right).

Explanation of the graph. On the left of Figure 6, all the bars are small and practically inside the bounds (lag 1: 0.011, lag 2: -0.038, lag 10: 0.031). The Ljung–Box test with 10 lags gives a p-value of 0.62, so there is no evidence of linear dependence between the returns. On the right, the bars for the squared returns are clearly positive and many exceed the bounds (lag 1: 0.113, lag 4: 0.075, lag 5: 0.070), and the Ljung–Box p-value is about 10^{-12}.

Conclusion. The returns are almost uncorrelated, which is compatible with an efficient market where the direction of the next move cannot be predicted. But they are not independent: their squares are positively correlated, which means that a day with a large move (up or down) tends to be followed by other days with large moves. This is volatility clustering, and it contradicts the constant-variance assumption of the random walk. Uncorrelated does not mean independent.

(c) Value-at-risk and expected shortfall

Objective. To measure the risk of holding \$1 000 000 of IBM for one day, and to see what the normal assumption does to this measure.

Method. The daily loss is minus the position times the return. The 99% value-at-risk (VaR) is the loss exceeded only once in 100 days, and the expected shortfall (ES) is the average loss on the days when the loss is beyond the VaR. Each is computed first from the historical returns (empirical quantile and average of the worst 1% of the days), then from a normal distribution with the sample mean (0.070%) and standard deviation (1.75%).

	Historical	Normal
99% VaR	\$42 954	\$40 003
99% ES	\$60 438	\$45 932

Conclusion. The historical method gives the larger risk for both measures. The difference is small for the VaR (+7%) but large for the expected shortfall (+32%). The reason is the heavy tails of real returns: the normal model describes correctly the point where the worst 1% begins, but it underestimates how bad the worst days are beyond that point. The VaR only looks at the threshold, while the ES looks at the whole tail, so the ES is the measure that reveals the problem of the normal model most clearly.

4 Exercise 4. Net, gross, log and multi-period returns (CRSPday and Garch)

(a) Net returns and log returns

Objective. To check how close $\log(1+R)$ is to R for daily data.

Method. The log returns are computed from the net returns, and the absolute difference between the two is taken day by day.

Series	Largest absolute difference	Mean absolute difference
GE	0.0030	0.000094
IBM	0.0077	0.000153
Mobil	0.0044	0.000083
CRSP	0.0023	0.000030

Conclusion. The approximation is very good for daily data: on average the difference is of the order of 10^{-4} or less. The largest difference occurs on the days of the largest moves (for IBM, the day with a return of +12.9% gives a log return of 12.2%), because the difference is about $R^2/2$, which is negligible for small returns but grows quickly for large ones. So the approximation is acceptable for daily data but should not be used for large returns or for long horizons.

(b) Wealth of \$1 invested on the first day

Objective. To show how returns accumulate by compounding.

Method. Starting from \$1, the wealth is multiplied each day by the gross return $1+R$ (cumulative product). The four wealth paths are drawn on a logarithmic vertical axis.

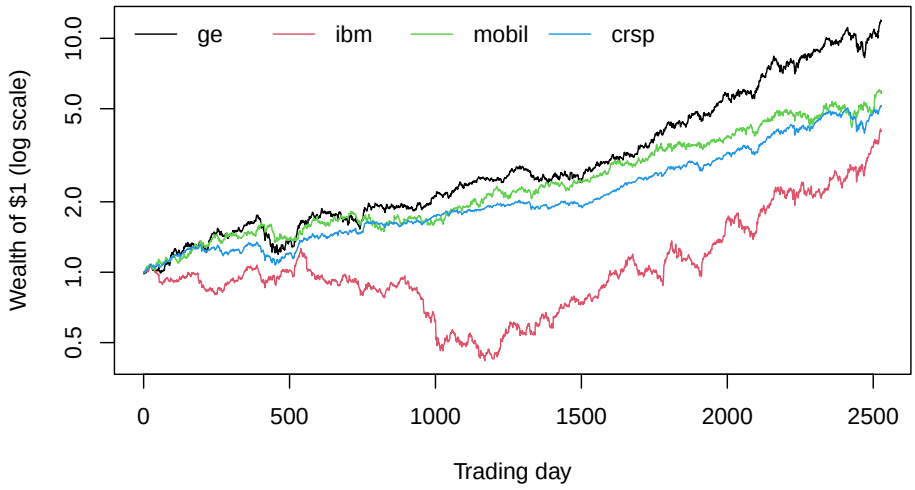


Figure 7: Wealth of $1 invested on the first day (log scale).

Results. After 2528 trading days (about ten years), the final wealth is $11.82 for GE, $5.80 for Mobil, $5.15 for CRSP and $3.99 for IBM.

Explanation of the graph. On a log scale, a constant growth rate gives a straight line, and the vertical distance between curves is the ratio of the wealths. GE rises almost steadily and ends far above the others, with an acceleration after 1995. Mobil and CRSP follow similar paths, and CRSP is smoother because it is diversified. The IBM curve is very different: it falls to a minimum of about $0.42 around 1993 (a loss of 58%) and then recovers to $3.99, which is a multiplication by more than 9 from the minimum. Even so, it ends as the worst of the four. This shows how a deep loss makes the recovery long.

(c) Monthly returns of GE in two ways

Objective. To verify the additive property of log returns over several periods.

Method. For each of the 120 months, the daily gross returns of GE are multiplied together and the logarithm of the result is taken. This is compared with the sum of the daily log returns in the same month.

Result. The two monthly log returns are the same: the largest difference over the 120 months is 10^{-16}, which is only the rounding error of the computer. This is the identity $\log \prod (1 + R_t) = \sum \log(1 + R_t)$. It is the main reason why log returns are used: the return over k periods is just the sum of the k one-period returns.

(d) Mean and variance of the monthly log return

Objective. To check the prediction of the random walk that the monthly log return has mean $k\mu$ and variance $k\sigma^2$.

Method. The daily mean and variance of the GE log returns are estimated ($\mu = 0.000977$, $\sigma^2 = 0.000188$, so $\sigma = 1.37\%$). A month has on average $k = 21.07$ trading days. The predicted

values are compared with the mean and variance of the 120 observed monthly log returns.

	Predicted ($k\mu$, $k\sigma^2$)	Observed
Mean	0.02058	0.02058
Variance	0.00395	0.00335

Comment. The means are identical. This is not a real test of the model: the sum of all the daily log returns is the same as the sum of all the monthly log returns, so the two averages must be equal by construction. The variance is more informative. The observed variance is about 15% smaller than predicted (a monthly standard deviation of 5.8% instead of 6.3%). Two explanations are possible. First, the daily returns of GE are slightly negatively autocorrelated (about -0.01, -0.02 and -0.03 at lags 1 to 3, and the Ljung–Box test with 10 lags rejects independence, with a p-value of 0.0099); when consecutive returns tend to offset each other, the variance of their sum is smaller than the sum of the variances. Second, the monthly variance is estimated from only 120 values, so it has a relative standard error of roughly 13%, and a part of the gap is simply sampling error. The random walk is therefore a reasonable but not perfect description at the monthly scale.

(e) Yearly net returns of GE

Method. The daily gross returns of each calendar year are multiplied together and 1 is subtracted.

Year	Net return	Year	Net return
1989	+48.7%	1994	+0.2%
1990	−8.1%	1995	+45.1%
1991	+37.4%	1996	+40.3%
1992	+15.1%	1997	+50.9%
1993	+26.0%	1998	+41.0%

Conclusion. The best year is 1997 (+50.9%), slightly ahead of 1989 (+48.7%), and the worst year is 1990 (−8.1%), which is the only year with a loss. The year 1994 is almost flat. These large positive values explain the final wealth of \$11.82 found in part (b).

(f) Exchange rate and its inverse (Garch)

Objective. To show why log returns are more convenient than net returns for exchange rates.

Method. The variable `dm` is the price of one German mark in dollars (1867 daily values). Its inverse is the price of one dollar in marks. For both quotations the daily net returns and the daily log returns are computed, and the return on the inverted quotation is compared with minus the return on the original quotation.

Results. For the log returns, the sum of the two series is zero for every day (largest absolute value 4×10^{-16}, i.e. computer rounding): the log return of the dollar in marks is exactly minus the log return of the mark in dollars. For the net returns, the sum is not zero: its largest absolute value is 0.0030, and it is exactly $R^2/(1+R)$, where R is the net return of the original quotation. The net returns do change sign every day, but their size is not the same. The averages show the consequence: the mean net return is positive for both quotations (0.0008% for the mark and

0.0052% for the dollar), as if both currencies appreciated against each other on average, while the mean log returns are -0.0022% and $+0.0022\%$, exactly opposite.

Conclusion. With log returns, the point of view does not matter: if the mark gains 1% in log terms, the dollar loses exactly 1% in log terms. With net returns the two views are not symmetric and the averages are misleading (both currencies seem to gain). For this reason, log returns are the natural convention for exchange rates.

5 Exercise 5. Adjustment for dividends (quantmod)

Objective. To measure how much of the return of a stock comes from dividends and to compare three ways of computing the total return.

Data. Daily prices and dividends of Coca-Cola (KO) from 2015 to 2024, downloaded from Yahoo Finance. This exercise needs an internet connection; the values below are those obtained when running the program for this TP.

Method. The total return over the whole period is computed in three ways: (1) from the closing prices only, as the last price divided by the first price minus 1; (2) from the closing prices plus the dividends, by multiplying the daily gross returns $(P_t + D_t)/P_{t-1}$, where D_t is the dividend paid on day t (zero on most days); (3) from the adjusted closing prices provided by Yahoo, which already include the dividends.

Method	Total return
Closing prices only	47.75%
Closing prices plus dividends	103.01%
Adjusted closing prices	103.05%

Conclusion. The dividends add 55.26 percentage points to the return: more than half of the total return of Coca-Cola over ten years (about 54%) comes from the dividends, and the price alone would give a very incomplete picture of the performance of the investor. This is typical of a mature company that pays high dividends. The return computed from the adjusted prices (103.05%) is very close to the dividend-adjusted return (103.01%); the remaining difference of 0.04 percentage points is small and is probably due to small differences in conventions, for example the exact day on which the dividend is taken into account (the ex-dividend date) and the rounding of the data provider. The conclusion is that adjusted prices can be used for return calculations, but raw closing prices should not be used for a long period for a stock that pays dividends.

6 Exercise 6. Geometric random walk (EuStockMarkets)

(a) Estimation of μ and σ

Objective. To calibrate the model on the DAX. The mean and the standard deviation of the 1859 daily log returns of the DAX are $\hat{\mu} = 0.000652$ (0.065% per day) and $\hat{\sigma} = 0.010301$ (1.03% per day, about 16.6% per year).

(b) Simulation of 1000 paths

Objective. To compare the real DAX with what the model would produce.

Method. Starting from the first DAX value, 1000 paths of the same length as the data (1860 values) are generated: each day the log price changes by a normal random number with mean $\hat{\mu}$ and standard deviation $\hat{\sigma}$, independent from the other days. Fifty paths are drawn together with the real DAX.

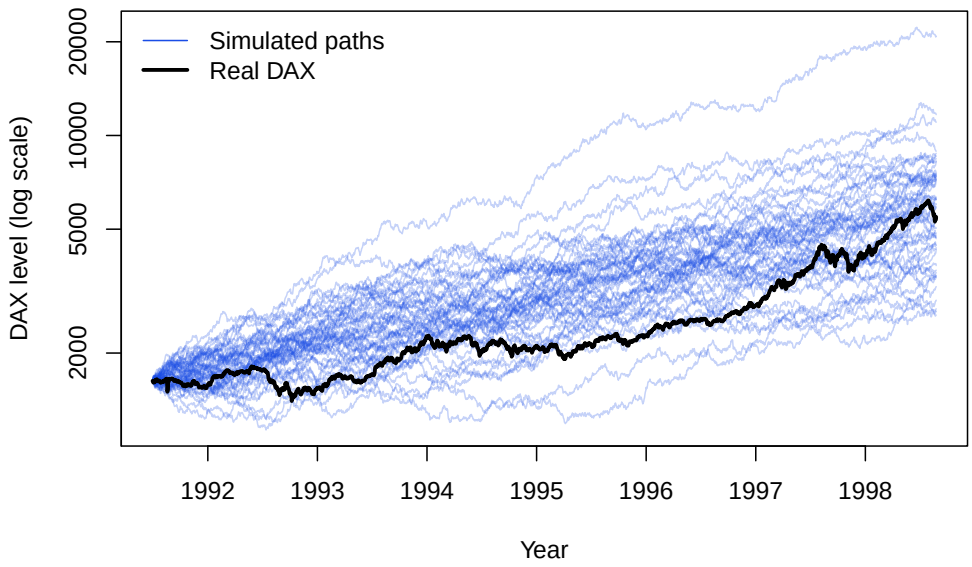


Figure 8: Fifty simulated paths of the geometric random walk (blue) and the real DAX (black), log scale.

Explanation of the graph. The blue paths in Figure 8 all start at the same point and spread out progressively, like a fan, because the uncertainty grows with the square root of time. At the end the simulated values are between about 1100 and 25 000 across the 1000 paths. The real DAX is a path that looks plausible inside the fan: it is in the lower part of the cloud from 1992 to 1996, then climbs fast in 1996–1997, and ends close to the middle of the cloud. Visually the real series is not different from a typical simulation, which shows that the geometric random walk reproduces the general behaviour of the price over several years. The simulated paths are smoother in appearance than the real one, which has clusters of large movements.

(c) Fraction of paths ending above the real final value

Result. The real final DAX value is 5473.7. In the simulation, 47.7% of the paths end above it. If the real DAX were a typical path of the model, this fraction would be about 50%, and 47.7% is very close to that. The real final value is a very ordinary outcome for the model. This is not surprising since $\hat{\mu}$ was estimated from the same data: the expected log price at the end is exactly the real final log price. The median of the simulated final values is 5366, just under the real value, and the mean is 5926, higher than the median, because the lognormal distribution is skewed to the right.

(d) Fraction of paths in the interval $\mu t \pm \sigma\sqrt{t}$

Objective. To verify that the simulation has the correct distribution at fixed dates: at time t the log price minus the starting log price is normal with mean μt and standard deviation $\sigma\sqrt{t}$, so the probability to be within one standard deviation of the mean is $\Phi(1) - \Phi(-1) = 68.27\%$.

Day t	Simulated fraction	Theoretical
100	65.8%	68.27%
500	68.8%	68.27%
1000	70.9%	68.27%
1500	67.7%	68.27%
1859	68.0%	68.27%

Conclusion. The simulated fractions are all between 65.8% and 70.9%, close to the theoretical 68.27%. With 1000 paths the sampling error is about ± 1.5 percentage points, which explains the small differences. The simulation is therefore correct, and the interval width grows as $\sqrt{t}$, as the fan shape of Figure 8 suggests.

(e) Autocorrelation of squared returns

Objective. To find which feature of the real data the model does not reproduce.

Method. The ACF of the squared log returns of the real DAX is compared with the ACF of the squared returns of one simulated path.

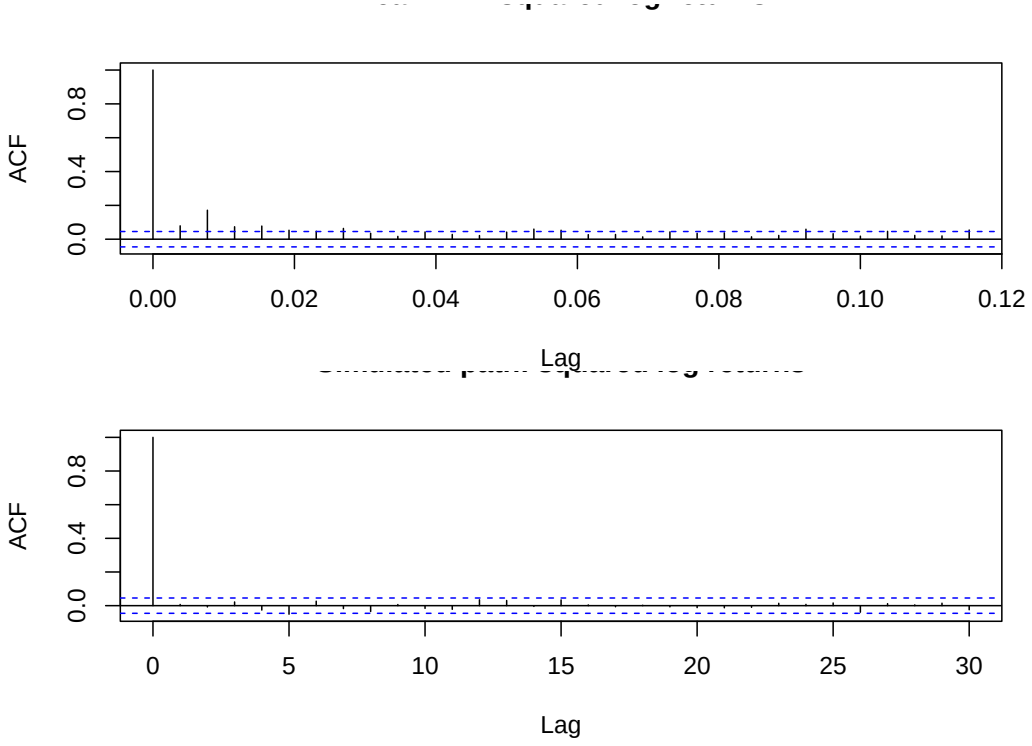


Figure 9: ACF of the squared returns: real DAX (top) and one simulated path (bottom).

Explanation of the graph. In the top panel of Figure 9, the real DAX has positive autocorrelations in squared returns at many lags, with values around 0.05 to 0.17 over the first five lags

(lag 2: 0.171), most of them above the dashed significance bound (about 0.045). In the bottom panel, the simulated path has autocorrelations of the squared returns that are all very small (-0.05 to 0.02 for the first five lags) and inside the bounds. The Ljung–Box test with 20 lags confirms it: the p-value is practically zero for the real DAX and 0.49 for the simulated path.

Conclusion. The model misses volatility clustering. In the real data, calm periods and turbulent periods follow one another, so the variance is not constant and large returns are followed by large returns. In the model each day is independent of the previous ones, with a constant variance, so no clustering can appear. This is the same weakness as the one found in Exercise 3(b).

7 Exercise 7. Leverage and Monte Carlo (CRSPday)

Problem. A hedge fund buys \$1 000 000 of the CRSP index with \$50 000 of its own capital and \$950 000 of borrowed money. If the value of the position closes below \$950 000, the fund must sell and loses all its capital.

(a) Estimation of μ and σ

From the 2528 daily log returns of the CRSP index, $\hat{\mu} = 0.000648$ (0.065% per day) and $\hat{\sigma} = 0.007776$ (0.78% per day). The sample skewness is -0.80 and the kurtosis is 11.2, so the returns are not normal.

(b) Probability of closing below \$950 000 within 45 days

Objective. To estimate the probability that the fund loses all its capital in 45 trading days.

Method. Two Monte Carlo simulations of 10 000 paths of 45 days each are done. In the first, the daily log returns are drawn from a normal distribution with the parameters of part (a). In the second (bootstrap), they are drawn with replacement from the 2528 historical log returns. For each path the value of the position is \$1 000 000 multiplied by the exponential of the cumulative log return, and the path is counted if it goes below \$950 000 on at least one day.

Results. The probability is 13.96% with normal returns and 13.71% with the bootstrap. So roughly one time in seven the fund is wiped out within two months, because a fall of only 5% is enough.

(c) Exit rule: sell above \$1 100 000, below \$950 000, or after 100 days

Objective. To evaluate a trading strategy with profit and loss targets.

Method. With 100-day paths, the fund sells at the first close above \$1 100 000 (a gain of \$100 000), at the first close below \$950 000, or at the 100th day otherwise. The profit is the value at exit minus the initial \$1 000 000. A profit of at least \$100 000 means that the upper level was reached, and a loss means a final value under \$1 000 000.

	Normal	Bootstrap
Probability of a profit of at least \$100 000	45.6%	45.6%
Probability of a loss	28.4%	27.8%
of which: closes below \$950 000 (capital lost)	22.6%	22.2%
Probability of exit after 100 days without hitting a level	31.9%	32.3%
Expected profit	\$46 670	\$47 076
Expected holding time	67.9 days	68.1 days

Interpretation. The strategy gains \$100 000 (twice the capital of the fund) in about 46% of the cases and loses in about 28%, including a loss of the whole capital in about 22%. The expected profit is positive, about \$47 000, which is almost equal to the initial capital of \$50 000, but this comes with a high risk. The position is held about 68 days on average.

(d) Normal versus bootstrap

Do the results agree? Yes, very closely: the differences are below one percentage point for all the probabilities, and below 1% for the expected profit and the holding time. The reason is that, over 45 or 100 days, the sum of the daily returns is close to normal by the central limit theorem, even if each daily return is not normal. What matters for the barriers is mainly the mean and the variance of the sum, and both methods use the same $\hat{\mu}$ and $\hat{\sigma}$.

Which one to trust? The bootstrap is more reliable in principle, because it does not assume a distribution: it keeps the skewness (-0.80) and the heavy tails (kurtosis 11.2) of the real returns, including the days of large losses. The normal model ignores them.

Does the bootstrap capture volatility clustering? No. The bootstrap draws the days independently, so it destroys the order of the data. Real CRSP returns have strong volatility clustering: the Ljung–Box test on the squared returns (20 lags) has a p-value below 10^{-15}. A simulated path from the bootstrap has the right distribution of daily returns but no periods of calm and turbulence. In real life, a loss of the capital is likely to occur during a turbulent period, so the probabilities of Exercise 7 may be underestimated by both methods. A better simulation would use a block bootstrap (which resamples blocks of consecutive days) or a model with time-varying variance, such as a GARCH model.

8 Exercise 8. Is the lognormal random walk a good model? (EuStockMarkets)

Objective. To test, for each of the four indices, the two assumptions of the model: that the log returns are normal, and that they are independent.

Method. For the 1859 daily log returns of each index:

- the skewness and the excess kurtosis (which are 0 for a normal distribution) are computed, and the Jarque–Bera statistic $JB = \frac{n}{6}\left(S^2 + \frac{K^2}{4}\right)$ is compared with a chi-square distribution with 2 degrees of freedom to test normality;
- the Ljung–Box test with 10 lags is applied to the log returns, to test for linear correlation, and to the squared log returns, to test for dependence in the variance. The 5% critical value of the statistic is 18.3.

Results.

	Skewness	Excess kurtosis	JB p-value	LB p-value returns	LB p-value squared returns
DAX	-0.554	6.28	$< 10^{-90}$	0.784	$< 10^{-10}$
SMI	-0.632	5.74	$< 10^{-90}$	0.254	$< 10^{-10}$
CAC	-0.177	2.39	$< 10^{-90}$	0.135	$< 10^{-10}$
FTSE	0.110	2.64	$< 10^{-90}$	0.0009	$< 10^{-10}$

The Jarque–Bera statistics are 3150 (DAX), 2672 (SMI), 450 (CAC) and 544 (FTSE), to be compared with the critical value of 5.99. The Ljung–Box statistics of the squared returns are 111, 98, 74 and 90, to be compared with 18.3.

Which assumptions are rejected?

- *Normality is rejected for all four markets*, with p-values that are essentially zero. The excess kurtosis is large and positive everywhere (from 2.4 to 6.3): the distributions have heavy tails. The skewness is negative for the DAX, SMI and CAC, which means that large losses are more frequent than large gains.
- *Independence of the returns themselves* is rejected only for the FTSE (p-value 0.0009); for the DAX, SMI and CAC there is no significant correlation between the returns.
- *Independence in a broader sense* is rejected for all four markets: the squared returns are strongly autocorrelated (p-values below 10^{-10}), which is volatility clustering.

So the assumption rejected for all four markets is the normality of the log returns, and also the independence (constant variance) as seen through the squared returns.

Is the model still useful? All models are false, but some models are useful. The lognormal random walk is false in the exact sense: returns have heavy tails, are slightly skewed, and their variance changes with time. But it is useful for several reasons. It is simple, with only two parameters, and the sum of log returns makes multi-period returns easy to handle. It reproduces the general shape of price paths and the growth of uncertainty with $\sqrt{t}$ (Exercise 6), and its predictions for Monte Carlo problems are close to the bootstrap ones (Exercise 7). It is also the starting point of the main pricing models in finance. However, it should not be trusted for what depends on the tails: probabilities of crashes (Exercise 2), value-at-risk and especially expected shortfall (Exercise 3), or risk in turbulent periods. For these, models with heavy tails (such as the t-distribution) and with a time-varying variance (such as GARCH) are needed.

General conclusion

The eight exercises give a consistent picture. Log returns are the natural tool: they add over time, they are symmetric for exchange rates, and they are close to net returns for daily data. Over the long term, prices follow a pattern that the geometric random walk reproduces well (the shape and the spread of the paths, the $\sqrt{t}$ rule, the correlation of the returns with their past which is almost zero). But three features of real returns are not in the model: heavy tails (Exercises 2, 3 and 8), skewness towards large losses (Exercises 2 and 8), and volatility clustering (Exercises 3, 6, 7 and 8). The practical consequence is that the model is a good first approximation for describing prices, but it underestimates extreme risk, and the tail risk must be handled with more realistic models.